EMA51005 - Discrete Mathematics

UNIT II - Combinatorics

Final Exam Answer Key

This answer key strictly follows the handwritten classwork flow: Induction Working Rule ($P(1) \to P(k) \to P(k+1)$), Generalised Pigeonhole Principle ($\left[\frac{m-n}{n}\right]+1$), step-by-step factorial expansion for ${}_nP_r$ and ${}_nC_r$, Inclusion–Exclusion, and characteristic root m for recurrence relations.

Part A - Short Answers

1. Assuming that repetitions are not permitted, how many four-digit numbers can be formed from the six digits $1, 2, 3, 5, 7, 8$?

Answer: The total number of given digits is $n = 6$ (namely $1, 2, 3, 5, 7, 8$), and we take $r = 4$ digits at a time without repetition.

The number of permutations of n things taken r at a time is given by:

$$nP_r = \frac{n!}{(n-r)!}$$

Substituting $n = 6$ and $r = 4$:

$$6P_4 = \frac{6!}{(6-4)!} = \frac{6!}{2!} = \frac{1 \times 2 \times 3 \times 4 \times 5 \times 6}{1 \times 2} = 3 \times 4 \times 5 \times 6 = \boxed{360}$$

2. Show that if seven colours are used to paint 50 bicycles, at least 8 bicycles will be the same colour.

Answer: Given:

$$m = 50 \quad [\text{Bicycles (Pigeons)}], \qquad n = 7 \quad [\text{Colours (Holes)}].$$

By the Generalised Pigeonhole Principle (as written in classwork):

$$= \left[\frac{m-n}{n}\right] + 1 = \left[\frac{50-7}{7}\right] + 1 = \left[\frac{43}{7}\right] + 1$$

$$= 6.14 + 1 \implies 7 + 1 = \boxed{8} \quad (\because \text{Greatest integer / ceiling step} : \lfloor (50-1)/7 \rfloor + 1 = 7$$

Hence, at least 8 bicycles will be of the same colour.

3. How many 7 letter words can be formed using the letters of the word BENZENE**?**

Answer: For the word BENZENE:

The total number of letters $= 7$

The letter E is repeated 3 times $= 3$

The letter N is repeated 2 times $= 2$

B $= 1$, Z $= 1$

Therefore, the number of distinct 7-letter words is:

$$= \frac{7!}{3! \times 2! \times 1! \times 1!} = \frac{1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7}{(1 \times 2 \times 3) \times (1 \times 2)}$$

$$= \frac{4 \times 5 \times 6 \times 7}{2} = 4 \times 5 \times 3 \times 7 = \boxed{420}$$

4. State Pigeonhole principle.

Answer: If $n + 1$ pigeons occupy n holes, then at least one hole has more than one pigeon.

5. Find the minimum number of students needed to guarantee that five of them belong to the same subject having major as English, Maths, Physics, and Chemistry.

Answer: Here, the number of subjects (holes) is $n = 4$ (English, Maths, Physics, Chemistry), and we require at least $k = 5$ students in the same subject.

By the Generalised Pigeonhole Principle:

$$\left\lfloor \frac{m-1}{n} \right\rfloor + 1 = 5 \quad \left(\text{or } \left\lceil \frac{m-n}{n} \right\rceil + 1 = 5 \right)$$

For the minimum number of students m:

$$\frac{m-1}{4} = 4 \implies m - 1 = 16 \implies m = \boxed{17}$$

6. Define the generalisation of the pigeonhole principle.

Answer: If m pigeons occupy n holes ($m > n$), then at least one hole contains at least:

$$\left\lceil \frac{m-n}{n} \right\rceil + 1 \quad \left(\text{i.e., } \left\lfloor \frac{m-1}{n} \right\rfloor + 1 \right) \text{ pigeons.}$$

7. A man hiked for 10 hours and covered a total distance of 45 km. It is known that he hiked 6 km in the first hour and only 3 km in the last hour. Show that he must have hiked at least 9 km within a certain period of 2 consecutive hours.

Answer: Total distance covered in 10 hours $= 45$ km. Distance covered in the 1st hour $= 6$ km and in the 10th hour $= 3$ km.

Distance covered in the remaining 8 hours (hours 2 to 9) is:

$$m = 45 - (6 + 3) = 45 - 9 = 36 \text{ km}.$$

Divide these 8 hours into $n = 4$ disjoint periods of 2 consecutive hours:
$(2, 3), (4, 5), (6, 7), (8, 9)$.

By the Generalised Pigeonhole Principle, at least one period of 2 consecutive hours must have a distance of at least:

$$\left\lceil \frac{m}{n} \right\rceil = \frac{36}{4} = \boxed{9 \text{ km}}$$

Hence, he must have hiked at least 9 km within a certain period of 2 consecutive hours.

8. Define Recurrence relation.

Answer: An equation that expresses the general term of a sequence $\{a_n\}$ in terms of one or more previous terms of the sequence, namely $a_1, a_2, \ldots, a_{n-1}$, for all positive integers n with $n \geq n_0$, where n_0 is a non-negative integer, is called a recurrence relation for the sequence $\{a_n\}$.

9. Write the general solution of recurrence relation.

Answer: For a linear recurrence relation, the general solution is the sum of the homogeneous solution $a_n^{(h)}$ and the particular solution $a_n^{(p)}$:

$$a_n = a_n^{(h)} + a_n^{(p)}.$$

For a second-order homogeneous recurrence relation with characteristic roots m_1, m_2:

If roots $m_1 \neq m_2$ are real and distinct: $y_n = Am_1^n + Bm_2^n$.

If roots $m_1 = m_2 = m$ are real and equal: $y_n = (A + Bn)m^n$.

10. Write the principle of Inclusion and Exclusion formula for the term $|A \cup B \cup C|$.

Answer: For any three finite sets A, B, C, the Principle of Inclusion and Exclusion formula is:

$$\boxed{|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|}$$

Part B - 6 Mark Answers

1. Use mathematical induction to prove that $n^3 + 2n$ is divisible by 3, for $n \geq 1$.

Answer: Let $P(n) : n^3 + 2n$ is divisible by 3.

Step 1: For n = 1

$$P(1) = 1^3 + 2(1) = 1 + 2 = 3,$$

which is divisible by 3. $\therefore P(1)$ is true.

Step 2: Assume P(k) is true

Assume $P(k)$ is true for $n = k$:

$$\frac{k^3 + 2k}{3} = c \implies k^3 + 2k = 3c \quad (\text{where } c \in \mathbb{Z}).$$

Step 3: To prove P(k+1) is true

For $n = k + 1$:

$$(k+1)^3 + 2(k+1) = (k^3 + 3k^2 + 3k + 1) + (2k + 2)$$

$$= (k^3 + 2k) + 3k^2 + 3k + 3$$

Using the inductive hypothesis $k^3 + 2k = 3c$:

$$= 3c + 3(k^2 + k + 1) = 3(c + k^2 + k + 1),$$

which is a multiple of 3, and hence divisible by 3.

$\therefore P(k+1)$ is true. Hence, by mathematical induction, $\boxed{n^3 + 2n \text{ is divisible by } 3}$ for all $n \geq 1$.

2. Prove by mathematical induction that $\frac{1}{1\cdot2}+\frac{1}{2\cdot3}+\frac{1}{3\cdot4}+\cdots+\frac{1}{n(n+1)}=\frac{n}{n+1}$.

Answer: Let $P(n):\frac{1}{1\cdot2}+\frac{1}{2\cdot3}+\cdots+\frac{1}{n(n+1)}=\frac{n}{n+1}$.

For n = 1:

$$P(1)=\frac{1}{1(2)}=\frac{1}{2},\qquad \text{RHS}=\frac{1}{1+1}=\frac{1}{2}.$$

$\therefore P(1)$ is true.

Assume for P(k):

$$\frac{1}{1\cdot2}+\frac{1}{2\cdot3}+\cdots+\frac{1}{k(k+1)}=\frac{k}{k+1}.$$

To prove P(k+1):

$$\frac{1}{1\cdot2}+\frac{1}{2\cdot3}+\cdots+\frac{1}{k(k+1)}+\frac{1}{(k+1)(k+2)}=\frac{k+1}{k+2}.$$

LHS (writing with previous term):

$$\text{LHS}=\frac{1}{1\cdot2}+\frac{1}{2\cdot3}+\cdots+\frac{1}{k(k+1)}+\frac{1}{(k+1)(k+2)}$$

$$=\frac{k}{k+1}+\frac{1}{(k+1)(k+2)}$$

$$=\frac{k(k+2)+1}{(k+1)(k+2)}=\frac{k^2+2k+1}{(k+1)(k+2)}$$

$$=\frac{(k+1)(k+1)}{(k+1)(k+2)}=\boxed{\frac{k+1}{k+2}}=\text{RHS}.$$

$\therefore P(k+1)$ is true. Hence proved by mathematical induction.

3. A club consisting of 6 men and 7 women, in how many ways can we select a committee of 4 persons which has at least one woman?

Answer: Given: Men $= 6$, Women $= 7$. A committee of 4 members is to be selected having at least one woman.

The possible ways of selection are:

1 Woman and 3 Men

2 Women and 2 Men

3 Women and 1 Man

4 Women and 0 Men

Therefore, the total number of ways is:

$$= {}^{7}C_1 \times {}^{6}C_3 + {}^{7}C_2 \times {}^{6}C_2 + {}^{7}C_3 \times {}^{6}C_1 + {}^{7}C_4 \times {}^{6}C_0$$

$$= \left(7 \times \frac{6 \times 5 \times 4}{1 \times 2 \times 3}\right) + \left(\frac{7 \times 6}{1 \times 2} \times \frac{6 \times 5}{1 \times 2}\right) + \left(\frac{7 \times 6 \times 5}{1 \times 2 \times 3} \times 6\right) + \left(\frac{7 \times 6 \times 5 \times 4}{1 \times 2 \times 3 \times 4} \times 1\right)$$

$$= (7 \times 20) + (21 \times 15) + (35 \times 6) + (35 \times 1)$$

$$= 140 + 315 + 210 + 35 = \boxed{700}$$

4. There are 250 students in an engineering college. Of these 188 have taken a course in Fortran, 100 have taken a course in C and 35 have taken courses in Java. Further 88 have taken courses in both Fortran and C. 23 have taken course in both C and Java and 29 have taken courses in both Fortran and Java. If 19 of these students have taken all the three courses, how many of these 250 students have not taken a course in any of these three programming languages?

Answer: Let A_1, A_2, A_3 denote the sets of students who have taken courses in Fortran, C, and Java respectively.

Given total students $n = 250$, and:

$$|A_1| = 188, \qquad |A_2| = 100, \qquad |A_3| = 35,$$

$$|A_1 \cap A_2| = 88, \qquad |A_2 \cap A_3| = 23, \qquad |A_1 \cap A_3| = 29, \qquad |A_1 \cap A_2 \cap A_3| = 19.$$

By the Principle of Inclusion and Exclusion:

$$|A_1 \cup A_2 \cup A_3| = |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| + |A_1 \cap A_2 \cap$$

$$= 188 + 100 + 35 - 88 - 29 - 23 + 19$$

$$= 323 - 140 + 19 = 202.$$

Therefore, the number of students who have not taken a course in any of these three programming languages is:

$$= n - |A_1 \cup A_2 \cup A_3| = 250 - 202 = \boxed{48}$$

5. Suppose there are 6 boys and 5 girls.

(i) In how many ways can they sit in a row.

(ii) In how many ways can they sit in row if the boys and girls each sit together.

(iii) In how many ways they can sit in a row if the girls are to sit together and the boys do not sit together.

Answer: Given: Number of boys $(\mathrm{B}) = 6$, Number of girls $(\mathrm{G}) = 5$, Total $= 11$.

(i) Number of ways they can sit in a row:

The number of boys : $6!$, and the number of girls : $5!$.

$$\text{The number of ways they can sit in a row is } = 11P_{11} = \frac{11!}{0!} = \boxed{11!} \quad (= 39,916,800).$$

(ii) If the boys and girls each sit together:

$$\text{The number of ways of B \& G sitting together } = 6! \times 5! \times 2!$$

$$= 720 \times 120 \times 2 = 720 \times 240 = \boxed{172800}$$

(iii) If the girls sit together and the boys do not sit together:

Treating the 5 girls as 1 unit, total ways with girls together is $7! \times 5! = 7 \times 6! \times 5!$, and subtracting the $2 \times 6! \times 5!$ ways where boys also sit together gives $5 \times 6! \times 5!$ (or placing the block of girls in the 5 internal gaps between the 6 boys, as written in classwork):

$$= 6! \times 5! \times 5$$

$$= 720 \times 120 \times 5 = 720 \times 600 = \boxed{432000}$$

6. Solve the recurrence relation $a_{n+2} - 6a_{n+1} + 9a_n = 3 \times 2^n + 7 \times 3^n$, $n \geq 0$, given that $a_0 = 1$ and $a_1 = 4$.

Answer: Given recurrence relation: $a_{n+2} - 6a_{n+1} + 9a_n = 3 \cdot 2^n + 7 \cdot 3^n$, with $a_0 = 1, a_1 = 4$.

Step 1: Homogeneous Solution

The characteristic equation is:

$$m^2 - 6m + 9 = 0 \implies (m-3)^2 = 0 \implies m = 3, 3.$$

$$a_n^{(h)} = (A + Bn)3^n.$$

Step 2: Particular Solution

For $3 \cdot 2^n$ (since 2 is not a root), let $a_n^{(p_1)} = C \cdot 2^n$:

$$C \cdot 2^{n+2} - 6C \cdot 2^{n+1} + 9C \cdot 2^n = 3 \cdot 2^n \implies C(4 - 12 + 9) = 3 \implies C = 3.$$

For $7 \cdot 3^n$ (since 3 is a double root), let $a_n^{(p_2)} = Dn^2 3^n$:

$$D(n+2)^2 3^{n+2} - 6D(n+1)^2 3^{n+1} + 9Dn^2 3^n = 7 \cdot 3^n$$

$$9D(n^2 + 4n + 4) - 18D(n^2 + 2n + 1) + 9Dn^2 = 7 \implies 18D = 7 \implies D = \frac{7}{18}.$$

Thus the general solution is:

$$a_n = (A + Bn)3^n + 3 \cdot 2^n + \frac{7}{18} n^2 3^n.$$

Step 3: Apply Initial Conditions (a_0 = 1, a_1 = 4)

$$n = 0: \quad a_0 = A + 3 = 1 \implies A = -2.$$

$$n = 1: \quad a_1 = (-2 + B)3 + 6 + \frac{7}{6} = 4 \implies 3B = 4 - \frac{7}{6} = \frac{17}{6} \implies B = \frac{17}{18}.$$

Hence the required solution is:

$$\boxed{a_n = \left(-2 + \frac{17}{18}n + \frac{7}{18}n^2\right)3^n + 3 \cdot 2^n}$$

7. How many positive integers n can be formed using the digits $3, 4, 4, 5, 5, 6, 7$, if n has to exceed $50,00,000$?

Answer: Given 7 digits: $3, 4, 4, 5, 5, 6, 7$, where 4 is repeated 2 times and 5 is repeated 2 times.

For a 7-digit integer n to exceed $50,00,000$, the first digit must be $5, 6$, or 7.

Case (i): First digit is 5

The remaining 6 digits are $\{3, 4, 4, 5, 6, 7\}$, where 4 is repeated 2 times:

$$\text{Number of ways} = \frac{6!}{2!} = \frac{720}{2} = 360.$$

Case (ii): First digit is 6

The remaining 6 digits are $\{3, 4, 4, 5, 5, 7\}$, where 4 is repeated 2 times and 5 is repeated 2 times:

$$\text{Number of ways} = \frac{6!}{2! \times 2!} = \frac{720}{4} = 180.$$

Case (iii): First digit is 7

The remaining 6 digits are $\{3, 4, 4, 5, 5, 6\}$, where 4 is repeated 2 times and 5 is repeated 2 times:

$$\text{Number of ways} = \frac{6!}{2! \times 2!} = \frac{720}{4} = 180.$$

Therefore, the total number of positive integers exceeding $50,00,000$ is:

$$= 360 + 180 + 180 = \boxed{720}$$

8. If 6 people A, B, C, D, E, F are seated about a round table, how many different circular arrangements are possible, if arrangements are considered the same when one can be obtained from the other by rotation? If A, B, C are females and the other are males, in how many arrangements do the genders alternate?

Answer: (i) Total circular arrangements of 6 people:

The number of circular arrangements of n persons around a round table is $(n-1)!$. For $n = 6$:

$$\text{Number of circular arrangements} = (6-1)! = 5! = 1 \times 2 \times 3 \times 4 \times 5 = \boxed{120}$$

(ii) Circular arrangements where genders alternate:

Given 3 females (A, B, C) and 3 males (D, E, F):

First, arrange the 3 females around the round table in $(3-1)! = 2!$ ways.

Next, arrange the 3 males in the 3 alternate seats between the females in $3!$ ways.

$$\text{Number of alternating arrangements} = (3-1)! \times 3! = 2! \times 3! = 2 \times 6 = \boxed{12}$$

9. A team of 11 players is to be chosen from 15 members. In how many ways can this be done if

(i) One particular player is always included.

(ii) Two such players have always to be included.

Answer: Total members $= 15$, Players to be chosen $= 11$.

(i) One particular player is always included:

$$(15-1)C_{(11-1)} \implies 14C_{10} = \frac{14!}{(14-10)!\,10!} = \frac{14!}{4!\,10!}$$

$$= \frac{11 \times 12 \times 13 \times 14}{1 \times 2 \times 3 \times 4} = 11 \times 13 \times 7 = 77 \times 13 = \boxed{1001}$$

(ii) Two such players have always to be included:

$$(15-2)C_{(11-2)} \implies 13C_{9} = \frac{13!}{(13-9)!\,9!} = \frac{13!}{4!\,9!}$$

$$= \frac{10 \times 11 \times 12 \times 13}{1 \times 2 \times 3 \times 4} = 5 \times 11 \times 13 = 55 \times 13 = \boxed{715}$$

10. Solve the recurrence relation $a_n - 2a_{n-1} = 3^n$, $a_1 = 5$.

Answer: Given recurrence relation: $a_n - 2a_{n-1} = 3^n$, with $a_1 = 5$.

Step 1: Homogeneous Solution

For $a_n - 2a_{n-1} = 0$, the characteristic equation is $m - 2 = 0 \implies m = 2$.

$$a_n^{(h)} = A \cdot 2^n.$$

Step 2: Particular Solution

Assume $a_n^{(p)} = C \cdot 3^n$. Substituting into the given relation:

$$C \cdot 3^n - 2C \cdot 3^{n-1} = 3^n$$

Dividing throughout by 3^{n-1}:

$$3C - 2C = 3 \implies C = 3.$$

$$\therefore a_n^{(p)} = 3 \cdot 3^n = 3^{n+1}.$$

Step 3: General Solution and Initial Condition

$$a_n = A \cdot 2^n + 3^{n+1}.$$

Putting $n = 1$ and using $a_1 = 5$:

$$a_1 = 2A + 3^2 = 5 \implies 2A + 9 = 5 \implies 2A = -4 \implies A = -2.$$

Therefore, the required solution is:

$$\boxed{a_n = 3^{n+1} - 2^{n+1}}$$

Part C - 10 Mark Answers

1. **Solve the recurrence relation $a_{n+1} - a_n = 3n^2 - n$; $n \geq 0$, $a_0 = 3$.**

Answer: Given recurrence relation: $a_{n+1} - a_n = 3n^2 - n$, $n \geq 0$, with $a_0 = 3$.

Step 1: Homogeneous Solution

The associated homogeneous relation is $a_{n+1} - a_n = 0$. Its characteristic equation is:

$$m - 1 = 0 \implies m = 1 \implies a_n^{(h)} = A(1)^n = A.$$

Step 2: Particular Solution

Since $m = 1$ is a root of multiplicity 1 and the RHS is a degree-2 polynomial, assume:

$$a_n^{(p)} = n(Bn^2 + Cn + D) = Bn^3 + Cn^2 + Dn.$$

Substituting into $a_{n+1} - a_n = 3n^2 - n$:

$$B\left[(n+1)^3 - n^3\right] + C\left[(n+1)^2 - n^2\right] + D\left[(n+1) - n\right] = 3n^2 - n$$

$$B(3n^2 + 3n + 1) + C(2n + 1) + D = 3n^2 - n$$

$$3Bn^2 + (3B + 2C)n + (B + C + D) = 3n^2 - n.$$

Comparing coefficients of like powers of n:

$$3B = 3 \implies B = 1,$$

$$3B + 2C = -1 \implies 3(1) + 2C = -1 \implies 2C = -4 \implies C = -2,$$

$$B + C + D = 0 \implies 1 - 2 + D = 0 \implies D = 1.$$

$$\therefore a_n^{(p)} = n^3 - 2n^2 + n = n(n-1)^2.$$

Step 3: General Solution and Initial Condition

$$a_n = A + n^3 - 2n^2 + n.$$

Using $a_0 = 3$ at $n = 0$: $a_0 = A + 0 = 3 \implies A = 3$. Hence:

$$\boxed{a_n = n^3 - 2n^2 + n + 3 = 3 + n(n-1)^2}$$

2. Use mathematical induction to prove that $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9, for $n \geq 1$.

Answer: Let $P(n) : n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9.

Step 1: For n = 1

$$P(1) = 1^3 + (1+1)^3 + (1+2)^3 = 1^3 + 2^3 + 3^3 = 1 + 8 + 27 = 36.$$

Since $36 = 9 \times 4$ is divisible by $9, \therefore P(1)$ is true.

Step 2: Assume P(k) is true

$$\frac{k^3 + (k+1)^3 + (k+2)^3}{9} = c \implies k^3 + (k+1)^3 + (k+2)^3 = 9c,$$

$$(k+1)^3 + (k+2)^3 = 9c - k^3.$$

Step 3: To prove P(k+1) is true

For $n = k+1$, consider:

$$(k+1)^3 + (k+2)^3 + (k+3)^3$$

Substituting $(k+1)^3 + (k+2)^3 = 9c - k^3$ and expanding $(k+3)^3$:

$$= (9c - k^3) + (k^3 + 9k^2 + 27k + 27)$$

$$= 9c + 9k^2 + 27k + 27 = 9(c + k^2 + 3k + 3),$$

which is divisible by $9. \therefore P(k+1)$ is true.

Hence, by mathematical induction, $\boxed{n^3 + (n+1)^3 + (n+2)^3 \text{ is divisible by } 9}$ for all $n \geq 1$.

3. Find a formula for the general term a_n of the Fibonacci sequence

$0, 1, 1, 2, 3, 5, 8, 13, \ldots$

Answer: Given the Fibonacci sequence $0, 1, 1, 2, 3, 5, 8, 13, \ldots$, the recurrence relation and initial conditions are:

$$a_n = a_{n-1} + a_{n-2} \implies a_n - a_{n-1} - a_{n-2} = 0 \quad (n \geq 2), \qquad a_0 = 0,\ a_1 = 1.$$

Step 1: Characteristic Equation

$$m^2 - m - 1 = 0 \implies m = \frac{1 \pm \sqrt{(-1)^2 - 4(1)(-1)}}{2} = \frac{1 \pm \sqrt{5}}{2}.$$

Let the two distinct roots be $m_1 = \frac{1+\sqrt{5}}{2}$ and $m_2 = \frac{1-\sqrt{5}}{2}$.

Step 2: General Solution

$$a_n = A\left(\frac{1+\sqrt{5}}{2}\right)^n + B\left(\frac{1-\sqrt{5}}{2}\right)^n.$$

Step 3: Apply Initial Conditions (a_0 = 0, a_1 = 1)

$$n = 0: \quad a_0 = A + B = 0 \implies B = -A.$$

$$n = 1: \quad a_1 = A\left(\frac{1+\sqrt{5}}{2}\right) - A\left(\frac{1-\sqrt{5}}{2}\right) = 1 \implies \sqrt{5}A = 1 \implies A = \frac{1}{\sqrt{5}}, \quad B =$$

Therefore, the general term a_n of the Fibonacci sequence is:

$$\boxed{a_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right], \quad n \geq 0}$$

4. Prove by mathematical induction that $1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$**.**

Answer: Let $P(n) : 1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$.

For n = 1:

$$P(1) = \frac{1}{3}(1)(2(1)-1)(2(1)+1) = \frac{1}{3}(2-1)(2+1) = \frac{1}{3}(1)(3) = 1.$$

$\therefore P(1)$ is true.

Assume P(k) is true:

$$1^2 + 3^2 + 5^2 + \cdots + (2k-1)^2 = \frac{1}{3}k(2k-1)(2k+1) = \frac{k(2k-1)(2k+1)}{3}.$$

To prove P(k+1) is true:

The next term is $(2(k+1)-1)^2 = (2k+1)^2$. Adding $(2k+1)^2$ on both sides:

$$\text{LHS} = \frac{k(2k-1)(2k+1)}{3} + (2k+1)^2$$

Factor out $(2k+1)$:

$$= (2k+1)\left[\frac{k(2k-1)}{3} + (2k+1)\right]$$

$$= (2k+1)\left[\frac{2k^2 - k + 6k + 3}{3}\right] = (2k+1)\left[\frac{2k^2 + 5k + 3}{3}\right]$$

$$= \boxed{\frac{(2k+1)(k+1)(2k+3)}{3}} = \text{RHS}.$$

$\therefore$ LHS $=$ RHS, so $P(k+1)$ is true. Hence proved by mathematical induction.

5. Use mathematical induction to show that $\frac{1}{\sqrt{1}}+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+\cdots+\frac{1}{\sqrt{n}}>\sqrt{n}$, for $n \geq 2$.

Answer: Let $P(n): \frac{1}{\sqrt{1}}+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+\cdots+\frac{1}{\sqrt{n}}>\sqrt{n}$.

Base Step:

For $n=2$: LHS $=\frac{1}{\sqrt{1}}+\frac{1}{\sqrt{2}}=1+0.707=1.707>\sqrt{2}\approx 1.414$. (In classwork: $P(1)=\frac{1}{\sqrt{1}}=1$). $\therefore P(n)$ is true for the base step.

Assume true for k:

$$P(k): \frac{1}{\sqrt{1}}+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+\cdots+\frac{1}{\sqrt{k}}>\sqrt{k}.$$

To prove P(k+1):

$$\frac{1}{\sqrt{1}}+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+\cdots+\frac{1}{\sqrt{k}}+\frac{1}{\sqrt{k+1}}>\sqrt{k+1}.$$

Consider the LHS and apply the inductive hypothesis (following the classwork steps):

$$\text{LHS}=\left(\frac{1}{\sqrt{1}}+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+\cdots+\frac{1}{\sqrt{k}}\right)+\frac{1}{\sqrt{k+1}}>\sqrt{k}+\frac{1}{\sqrt{k+1}}$$

$$=\frac{\sqrt{k}\sqrt{k+1}+1}{\sqrt{k+1}}>\frac{\sqrt{k}\sqrt{k}+1}{\sqrt{k+1}}=\frac{k+1}{\sqrt{k+1}}$$

$$=\frac{\sqrt{k+1}\sqrt{k+1}}{\sqrt{k+1}}=\boxed{\sqrt{k+1}}.$$

$\therefore P(k+1)$ is true. Hence proved by mathematical induction.

6. Show that $a^n - b^n$ is divisible by $(a-b)$ for all $n, n \geq 1$ using principle of mathematical induction.

Answer: Let $P(n) : a^n - b^n$ is divisible by $(a-b)$.

Step 1: For n = 1

$$P(1) = a^1 - b^1 = a - b,$$

which is divisible by $(a-b)$. $\therefore P(1)$ is true.

Step 2: Assume P(k) is true

$$\frac{a^k - b^k}{a-b} = c \implies a^k - b^k = c(a-b) \implies a^k = c(a-b) + b^k.$$

Step 3: To prove P(k+1) is true

For $n = k+1$, consider:

$$a^{k+1} - b^{k+1} = a^k \cdot a^1 - b^k \cdot b^1$$

Substituting $a^k = c(a-b) + b^k$:

$$= \left[c(a-b) + b^k\right] a - b^k \cdot b$$

$$= ac(a-b) + ab^k - b \cdot b^k = ac(a-b) + b^k(a-b)$$

$$= (a-b)(ac + b^k),$$

which has $(a-b)$ as a factor and is therefore divisible by $(a-b)$.

$\therefore P(k+1)$ is true. Hence $\boxed{a^n - b^n \text{ is divisible by } (a-b)}$ for all $n \geq 1$.

7. Find the number of integers between 1 and 250 both inclusive that are not divisible by any of the integers $2, 3, 5$ and 7.

Answer: Let A_1, A_2, A_3, A_4 denote the sets of integers between 1 and 250 divisible by $2, 3, 5, 7$ respectively.

Step 1: Cardinalities of Single Sets

$$|A_1| = \left\lfloor \frac{250}{2} \right\rfloor = 125, \quad |A_2| = \left\lfloor \frac{250}{3} \right\rfloor = 83, \quad |A_3| = \left\lfloor \frac{250}{5} \right\rfloor = 50, \quad |A_4| = \left\lfloor \frac{250}{7} \right\rfloor =$$

Step 2: Cardinalities of Pairwise Intersections

$$|A_1 \cap A_2| = \left\lfloor \frac{250}{2 \times 3} \right\rfloor = \left\lfloor \frac{250}{6} \right\rfloor = 41, \qquad |A_1 \cap A_3| = \left\lfloor \frac{250}{2 \times 5} \right\rfloor = \left\lfloor \frac{250}{10} \right\rfloor = 25,$$

$$|A_1 \cap A_4| = \left\lfloor \frac{250}{2 \times 7} \right\rfloor = \left\lfloor \frac{250}{14} \right\rfloor = 17, \qquad |A_2 \cap A_3| = \left\lfloor \frac{250}{3 \times 5} \right\rfloor = \left\lfloor \frac{250}{15} \right\rfloor = 16,$$

$$|A_2 \cap A_4| = \left\lfloor \frac{250}{3 \times 7} \right\rfloor = \left\lfloor \frac{250}{21} \right\rfloor = 11, \qquad |A_3 \cap A_4| = \left\lfloor \frac{250}{5 \times 7} \right\rfloor = \left\lfloor \frac{250}{35} \right\rfloor = 7.$$

Step 3: Cardinalities of Three-Way and Four-Way Intersections

$$|A_1 \cap A_2 \cap A_3| = \left\lfloor \frac{250}{2 \times 3 \times 5} \right\rfloor = 8, \qquad |A_1 \cap A_2 \cap A_4| = \left\lfloor \frac{250}{2 \times 3 \times 7} \right\rfloor = 5,$$

$$|A_2 \cap A_3 \cap A_4| = \left\lfloor \frac{250}{3 \times 5 \times 7} \right\rfloor = 2, \qquad |A_1 \cap A_3 \cap A_4| = \left\lfloor \frac{250}{2 \times 5 \times 7} \right\rfloor = 3,$$

$$|A_1 \cap A_2 \cap A_3 \cap A_4| = \left\lfloor \frac{250}{2 \times 3 \times 5 \times 7} \right\rfloor = \left\lfloor \frac{250}{210} \right\rfloor = 1.$$

Step 4: Principle of Inclusion and Exclusion

$$|A_1 \cup A_2 \cup A_3 \cup A_4| = (125 + 83 + 50 + 35) - (41 + 25 + 17 + 16 + 11 + 7) + (8 + 5 +$$

$$= 293 - 117 + 18 - 1 = \boxed{193}$$

Therefore, the number of integers between 1 and 250 not divisible by any of $2, 3, 5, 7$ is:

$$= 250 - |A_1 \cup A_2 \cup A_3 \cup A_4| = 250 - 193 = \boxed{57}$$

8. How many bit strings of length 10 contain

(a) Exactly four 1's

(b) Atmost four 1's

(c) At least four 1's

(d) An equal number of 0's and 1's ?

Answer: Given: The bit string length is 10.

(a) Exactly four 1's:

$$= 10C_4 = \frac{10!}{4!\,6!} = \frac{7 \times 8 \times 9 \times 10}{1 \times 2 \times 3 \times 4} = 30 \times 7 = \boxed{210}$$

(b) At most four 1's (i.e., 0, 1, 2, 3, or 4 ones):

$$= \frac{10!}{0!\,10!} + \frac{10!}{1!\,9!} + \frac{10!}{2!\,8!} + \frac{10!}{3!\,7!} + \frac{10!}{4!\,6!}$$

$$= 1 + 10 + 45 + 120 + 210 = \boxed{386}$$

(c) At least four 1's (i.e., 4, 5, 6, 7, 8, 9, or 10 ones):

4 1's and 6 0's, 5 1's and 5 0's, 6 1's and 4 0's, 7 1's and 3 0's

8 1's and 2 0's, 9 1's and 1 0's, 10 1's and 0 0's

$$= \frac{10!}{4!\,6!} + \frac{10!}{5!\,5!} + \frac{10!}{6!\,4!} + \frac{10!}{7!\,3!} + \frac{10!}{8!\,2!} + \frac{10!}{9!\,1!} + \frac{10!}{10!\,0!}$$

$$= 210 + 252 + 210 + 120 + 45 + 10 + 1 = \boxed{848}$$

(d) An equal number of 0's and 1's (five 0's and five 1's):

$$= \frac{10!}{5!\,5!} = \frac{6 \times 7 \times 8 \times 9 \times 10}{1 \times 2 \times 3 \times 4 \times 5} = \boxed{252}$$

9. Solve the recurrence relation $a_n = 2(a_{n-1} - a_{n-2})$, $n \geq 2$ and $a_0 = 1, a_1 = 2$.

Answer: Given recurrence relation: $a_n = 2(a_{n-1} - a_{n-2}) \implies a_n - 2a_{n-1} + 2a_{n-2} = 0$, with $a_0 = 1, a_1 = 2$.

Step 1: Characteristic Equation

$$m^2 - 2m + 2 = 0 \implies m = \frac{2 \pm \sqrt{4-8}}{2} = 1 \pm i.$$

Step 2: General Solution (Algebraic Form)

$$a_n = A(1+i)^n + B(1-i)^n.$$

Applying initial conditions $a_0 = 1$ and $a_1 = 2$:

$$n = 0: \quad A + B = 1,$$

$$n = 1: \quad A(1+i) + B(1-i) = 2 \implies (A+B) + i(A-B) = 2 \implies 1 + i(A-B) =$$

Solving for A and B gives $A = \frac{1-i}{2}$ and $B = \frac{1+i}{2}$. Thus:

$$a_n = \left(\frac{1-i}{2}\right)(1+i)^n + \left(\frac{1+i}{2}\right)(1-i)^n = (1+i)^{n-1} + (1-i)^{n-1}.$$

Step 3: Polar / Trigonometric Form

Writing $1 \pm i = \sqrt{2}\left(\cos\frac{\pi}{4} \pm i\sin\frac{\pi}{4}\right)$, we have $a_n = (\sqrt{2})^n\left(c_1\cos\frac{n\pi}{4} + c_2\sin\frac{n\pi}{4}\right)$.

Using $a_0 = 1 \implies c_1 = 1$, and $a_1 = 2 \implies \sqrt{2}\left(\frac{1}{\sqrt{2}} + \frac{c_2}{\sqrt{2}}\right) = 2 \implies c_2 = 1$:

$$\boxed{a_n = (\sqrt{2})^n\left(\cos\frac{n\pi}{4} + \sin\frac{n\pi}{4}\right)}$$

10. In how many ways can 20 students out of a class of 30 be selected for an extra-curricular activity, if

(a) Rama refuses to be selected ?

(b) Raja insists on being selected ?

(c) Gopal and Govind insist on being selected ?

(d) Either Gopal or Govind or both get selected ?

Answer: Total students in class $= 30$, Students to be selected $= 20$.

(a) Rama refuses to be selected (choose 20 from remaining 29):

$$= 29C_{20} = \frac{29!}{(29-20)!\,20!} = \frac{29!}{9!\,20!}$$

$$= \frac{21 \times 22 \times 23 \times 24 \times 25 \times 26 \times 27 \times 28 \times 29}{1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \times 9}$$

$$= 11 \times 23 \times 6 \times 5 \times 13 \times 7 \times 29 = \boxed{10015005}$$

(b) Raja insists on being selected (choose 19 from remaining 29):

$$= (30-1)C_{(20-1)} = 29C_{19} = \frac{29!}{10!\,19!} = \boxed{20030010}$$

(c) Gopal and Govind insist on being selected (choose 18 from remaining 28):

$$= (30-2)C_{(20-2)} = 28C_{18} = \frac{28!}{10!\,18!} = \boxed{13123110}$$

(d) Either Gopal or Govind or both get selected:

Classwork notebook step ($30C_{20} - 28C_{18}$):

$$30C_{20} - 28C_{18} = 30045015 - 13123110 = \boxed{16921905}$$

By complement / inclusion-exclusion ($30C_{20} - 28C_{20} = 28C_{19} + 28C_{19} + 28C_{18}$):

$$30C_{20} - 28C_{20} = 30045015 - 3108105 = \boxed{26936910}$$